

# HUSSEIN ELSAYED

Unity Developer | XR / VR | Multiplayer | EdTech

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Open to remote, hybrid & on-site roles internationally · Available to relocate

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YouTube: <https://www.youtube.com/@egygamesstudios>

Portfolio: <https://husseinelayed18.github.io/>

## PROFESSIONAL SUMMARY

Unity Developer with professional experience building **multiplayer games, VR/XR applications, AR experiences, robotics simulations, educational technology platforms, interactive training systems, and mobile applications**. Strong experience in **C#, Unity, multiplayer and client-server architecture, VR/XR, AR, UI/UX, backend integration, REST APIs, visual programming, custom editors, Python code generation, AI/computer vision, and third-party SDK integration**. Experienced in developing complete interactive systems from **software architecture and UI/UX design to gameplay, networking, authentication, platform services, and performance optimization**. Educational technology experience includes developing robotics programming and simulation systems and teaching programming through **Scratch, Blockly, Python, JavaScript, React, Flutter, C#, Unity, and robotics**.

## PROFESSIONAL EXPERIENCE

### Kid Koder Academy

**Instructor — Programming, Game Development & Robotics**

**Aug 2026 – Present**

- Teach programming **from scratch**, starting with computational thinking and beginner-friendly visual programming.
- Teach **Scratch and Blockly**, including variables, conditions, loops, events, operators, functions, and programming logic.
- Teach students how **Scratch-style block editors work**, including concepts behind visual programming and custom block-based programming environments.
- Guide students through building games from scratch, including **Pong, multiplayer Pong, Super Mario-style games, and car/racing games**.
- Use game development to teach programming, problem-solving, algorithms, and logical thinking.
- Progress students from visual programming into text-based programming, including **Python and JavaScript**.
- Introduce modern development technologies through practical projects, including **React and Flutter**.
- Teach **C# and Unity** for students progressing toward professional game development.
- Teach robotics and programming concepts through hands-on projects and interactive simulations.
- Adapt lessons for students with different programming experience levels.

### Freelance / Independent Unity Developer - [EgyGames](#)

**2026 – Present**

#### *Sigato V2 — Multiplayer 2D Board Game*

- Continued development of **Sigato V2**, expanding the original multiplayer game developed at DGSs.
- Designed and implemented the game's **UI/UX independently using Figma**, translating designs into Unity UI.
- Designed and implemented the project's software architecture using **Extenject / Dependency Injection**.
- Extended multiplayer architecture with **client-server synchronization, online communication, lobby, and matchmaking systems**.
- Integrated **Photon PUN 2** for multiplayer networking.
- Integrated **Unity Gaming Services (UGS), Firebase, and Google Sign-In**.

- Integrated **Google Play Games and Game Center**.
- Implemented **Analytics, In-App Purchases, Leaderboards, Achievements, Advertising, and localization**.
- Implemented **English, Arabic, and French localization**.
- Developed **daily rewards, challenges, shop, rewards, themes, audio, settings, and reusable popup systems**.
- Used **UniRx** and **LitMotion** for reactive systems and responsive UI interactions.
- Optimized the application for **mobile and cross-platform deployment**.

EKSON Technology / AlMotahida — [SciVR](#)

#### Unity Developer — VR / XR / Educational Applications

Sep 2024 – Jul 2026

- Developed **VR/XR, educational, medical, scientific, and interactive Unity applications**.
- Developed multiple VR projects using **VRIF (VR Interaction Framework)**.
- Built and optimized VR applications for **Meta Quest and Pico** standalone headsets.
- Developed and deployed **WebGL** interactive experiences for browser-based platforms.
- Implemented immersive VR interactions, object manipulation, UI, navigation, animation, audio, and interactive workflows.
- Developed VR virtual tours including **Pizza Tower and Acropolis** experiences.
- Developed medical and dental VR applications including **Dentist, Dentist Hygiene, Skin Care**, and related training experiences.
- Contributed to **5+ Mathematics, 5+ Biology, and 5+ CTE** educational projects.
- Optimized applications for standalone VR hardware and WebGL deployment.

DGSs

#### Game Developer — Sigato V1

Jul 2024 – Aug 2024

- Programmed and launched **Sigato**, a multiplayer 2D board game deployed directly to the Google Play Store.
- Implemented **real-time client-server synchronization**.
- Built **lobby and matchmaking pipelines**.
- Implemented online communication states and multiplayer gameplay synchronization.
- Established the initial gameplay, UI, and networking architecture that became the foundation for **Sigato V2**.

Thaka (Thaka international) / BE-STEAM (prev name) — [01Skills](#)

#### Unity Developer / Robotics & EdTech Developer

2023 – 2024

- Contributed to [01Skills](#), an educational technology platform serving thousands of students/users through interactive coding, robotics simulation, learning, practice, and competition.
- Developed and enhanced robotics simulation supporting **LEGO EV3 and SPIKE-style programming**.
- Extended block-based programming with **loops, conditions, logic, comparisons, arithmetic, variables, robot movement, sensors, and control structures**.
- Implemented support for **complex nested mathematical and logical expressions**.
- Developed comprehensive **EV3 and SPIKE block-programming systems**.
- Built a custom programming editor supporting **Block-to-Python and Code-to-Blocks** workflows.
- Implemented project loading and saving and a portable **Python code-generation workflow**.
- Connected programming blocks with simulated robot behavior, robotics mission logic, and challenges.
- Worked on the **Panama educational robotics mission**, including robot simulation, block programming, mission objectives, and execution logic.

PSTC — Saudi Arabia

#### Game Developer

Oct 2021 – Apr 2022

- Engineered a **multiplayer gamified mobile application** integrated into the InsuranceMarket platform.
- Implemented real-time multiplayer functionality using the **AppWarp multiplayer networking plugin**.

- Built multiplayer rooms, real-time communication, and synchronized game states.
- Built secure **login/sign-up authentication, user profiles, guest sessions, and real-time live chat rooms.**
- Integrated **Unity Ads and Start.io** SDK networks.
- Performed **texture compression, script refactoring, and code profiling** to improve application size and runtime performance.
- Optimized assets, scripts, rendering, and runtime performance for mobile devices.

## FREELANCE PROJECTS

### VR Clean Desk Policy

#### Interactive VR Training Simulation

- Developed a Unity-based VR workplace training simulation.
- Implemented **VR interactions, object manipulation, task-based workflows, and spatial audio/narration.**
- Developed the experience for standalone VR platforms.

### GetMoney

#### Economic Simulation Game — Unity / PHP / MySQL

- Developed gameplay systems using **Unity, PHP, and MySQL.**
- Built custom backend functionality through **REST APIs.**
- Connected Unity gameplay systems with server-side data.
- Implemented interactive economic and simulation mechanics.

## SELECTED PROJECTS

### AR Card Gameplay System

#### Augmented Reality / Unity

- Developed an AR tracking system mapping **physical cards and anchors to virtual interfaces and gameplay elements.**
- Implemented image/marker tracking to recognize physical cards and connect them with corresponding digital interactions.
- Created an interactive bridge between **physical objects and virtual Unity gameplay/UI systems.**

### AI & Computer Vision

- Developed Unity-based AI and computer vision experiences using **Unity Sentis, Unity ML-Agents, AI Behavior Trees, and object detection/recognition.**

### Cinema Cars Race

- Developed a Unity racing experience involving gameplay systems, vehicles, UI, and interactive environments.

### Unity Matching Game

- Developed a matching/puzzle game with gameplay, UI, progression, rewards, and interactive systems.

## TECHNICAL SKILLS

### Programming Languages

**C#, Python, Java, C++, JavaScript, PHP, HTML5**

### Unity & Game Development

**Unity, Unity 6, 2D, 3D, Gameplay Programming, UI/UX, Physics, Animation, Simulation, Custom Editors**

## Multiplayer & Networking

**Client-Server Architecture, Multiplayer, Photon PUN 2, AppWarp, Mirror, WebSockets, Lobby Systems, Matchmaking, Real-Time Synchronization**

## XR / VR / AR

**VR, XR, AR, VRIF, OpenXR, XR Interaction Toolkit, Meta Quest, Pico, Vuforia, VR Training, Virtual Tours**

## Backend & APIs

**PHP, REST APIs, GraphQL, Firebase, MySQL, MongoDB, MongoDB Realm**

## Architecture & Software Engineering

**OOP, SOLID, Design Patterns, Dependency Injection, Extenject, Event-Driven Architecture, Service Locator, Modular Systems, Data Structures & Algorithms, Performance Optimization**

## UI/UX & Design

**Figma, Unity UI, UI/UX Design, Prototyping, Responsive UI**

## Visual Programming & Robotics

**Scratch, Blockly, Block-Based Programming, LEGO EV3, LEGO SPIKE, Visual Programming, Python Code Generation, Block-to-Code, Code-to-Blocks, Compiler Pipelines, Custom Programming Editors**

## AI & Computer Vision

**Unity Sentis, Unity ML-Agents, AI Behavior Trees, Computer Vision, Object Detection**

## Platform Services & SDKs

**Firebase Authentication, Google Sign-In, Analytics, In-App Purchases, Unity Ads, Start.io, Leaderboards, Achievements, Google Play Games, Game Center**

## Other Technologies

**React, Flutter**

## Platforms

**Android, iOS, PC, WebGL, Mobile, Meta Quest, Pico**

## Tools

**Git, GitHub, Jira, Visual Studio, Android Studio, Blender, Photoshop, Figma**

## LANGUAGES

**Arabic:** Native

**English:** Professional Working Proficiency